

SHOEBURYNESSE LANDWICK, United Kingdom

WMO: 036930

Lat: 51.5547N

Lon: 0.8269E

Elev: 10

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	29.8	22.6	17.0	31.7	25.0	19.1	33.1	31.0	47.9	27.9	46.1	7.9	310	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	77.3	65.7	74.7	64.4	72.4	63.2	67.7	74.8	66.1	72.3	64.7	70.5	10.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.9	92.2	71.0	63.6	87.9	69.6	62.1	83.4	68.1	32.0	74.9	30.8	72.4	29.7	70.6	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	22.7	20.4	DB	23.2	84.6	3.1	2.7	21.0	86.5	19.2	88.1	17.4	89.6	15.2	91.6	(4)
(5)			WB	22.6	71.1	3.0	1.7	20.5	72.3	18.8	73.3	17.1	74.2	14.9	75.4	(5)

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	41.2	41.6	44.1	48.4	53.9	59.3	63.9	64.0	59.4	53.9	46.9	42.5	(6)
(7)		DBStd	9.67	5.71	5.56	5.11	5.22	4.90	4.52	4.11	3.99	4.26	4.96	5.59	6.13	(7)
(8)		HDD50	1195	274	237	190	89	20	1	0	0	0	20	123	241	(8)
(9)		HDD65	4952	737	654	648	498	343	177	70	66	173	344	544	698	(9)
(10)		CDD50	1804	3	3	8	41	142	281	431	434	283	141	29	8	(10)
(11)		CDD65	85	0	0	0	0	0	7	36	36	6	0	0	0	(11)
(12)		CDH74	313	0	0	0	0	3	30	137	124	16	3	0	0	(12)
(13)		CDH80	32	0	0	0	0	0	2	15	15	0	0	0	0	(13)
(14)	Wind	WSAvg	10.5	11.8	11.7	11.0	10.2	10.2	9.6	9.1	9.5	10.0	10.7	10.6	11.3	(14)
(15)	Precipitation	PrecAvg	19.8	1.5	1.1	1.4	1.4	1.4	1.9	1.6	1.9	1.9	2.0	2.0	1.8	(15)
(16)		PrecMax	25.9	4.8	2.4	3.3	2.6	2.6	4.3	4.1	4.3	6.5	5.1	4.0	3.7	(16)
(17)		PrecMin	13.0	0.4	0.2	0.1	0.2	0.0	0.2	0.3	0.4	0.1	0.1	0.4	0.5	(17)
(18)		PrecStd	2.9	0.8	0.6	0.9	0.7	0.7	1.1	0.9	0.9	1.5	1.3	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	56.7	59.6	68.0	72.4	77.9	81.9	81.9	76.0	68.7	61.1	57.1	(19)
(20)		MCWB	52.3	51.6	51.9	58.3	62.1	66.2	66.7	68.0	65.5	61.6	57.8	54.1		(20)
(21)		2%	DB	53.4	53.6	56.4	62.7	67.8	73.6	77.4	77.1	71.9	65.0	58.5	55.0	(21)
(22)		MCWB	50.3	49.9	50.3	54.7	59.6	63.3	65.6	65.9	63.5	59.7	55.3	52.2		(22)
(23)		5%	DB	51.5	51.7	54.2	59.3	65.2	70.5	74.7	74.3	69.0	62.8	56.8	53.4	(23)
(24)		MCWB	48.9	48.0	49.1	52.5	57.5	61.7	64.3	64.3	61.3	58.2	53.7	50.9		(24)
(25)		10%	DB	49.6	49.8	52.3	56.9	62.7	67.8	72.5	71.8	66.8	61.1	54.9	51.6	(25)
(26)		MCWB	47.2	46.6	47.9	51.1	55.9	60.0	63.2	62.8	60.0	57.3	52.1	49.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	52.4	53.4	59.2	63.7	68.0	69.6	69.6	67.6	62.9	58.4	54.6	(27)
(28)		MADB	54.8	55.5	57.3	66.8	69.9	76.0	78.2	78.6	73.0	67.1	60.4	56.3		(28)
(29)		2%	WB	51.0	50.5	51.7	55.9	60.9	65.1	67.6	67.7	64.9	60.9	56.1	52.9	(29)
(30)		MADB	52.8	53.1	55.0	61.0	66.3	71.3	74.9	74.6	70.0	63.9	58.0	54.7		(30)
(31)		5%	WB	49.2	48.7	50.2	53.7	58.8	63.1	66.0	66.1	63.1	59.5	54.3	51.3	(31)
(32)		MADB	51.2	51.0	53.3	57.9	63.6	68.6	72.6	71.9	67.5	62.1	56.3	53.2		(32)
(33)		10%	WB	47.4	47.1	48.5	52.0	57.1	61.3	64.5	64.5	61.3	58.1	52.5	49.4	(33)
(34)		MADB	49.4	49.5	51.6	56.1	61.6	66.3	70.7	70.0	65.4	60.5	54.6	51.4		(34)
(35)	Mean Daily Temperature Range	MDBR	8.3	9.5	11.4	13.5	13.8	14.1	14.5	14.3	13.3	11.0	9.3	8.1		(35)
(36)		5% DB	MCDBR	9.4	11.0	13.9	16.6	17.5	17.7	18.1	17.4	15.5	11.9	9.0	8.0	(36)
(37)		MCWBR	8.9	8.8	9.6	10.0	10.3	9.9	8.9	8.5	8.6	8.3	7.7	7.4		(37)
(38)		5% WB	MCDBR	9.3	10.3	12.9	15.9	16.5	16.4	16.8	16.0	14.3	10.9	8.7	8.3	(38)
(39)		MCWBR	9.1	8.8	9.7	10.4	10.7	10.1	9.1	8.6	9.0	8.4	8.1	8.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.333	0.350	0.387	0.408	0.414	0.415	0.413	0.404	0.383	0.366	0.339	0.324		(40)
(41)		taud	2.374	2.377	2.267	2.240	2.268	2.289	2.307	2.353	2.402	2.425	2.416	2.365		(41)
(42)		Ebn at Noon	213	243	253	263	267	267	266	262	254	234	209	194		(42)
(43)		Edh at Noon	20	26	34	40	40	40	38	35	30	24	19	17		(43)
(44)	All-Sky Solar Radiation	RadAvg	247	459	815	1308	1556	1725	1637	1365	1024	586	298	191		(44)
(45)		RadStd	23	69	101	124	112	134	146	105	82	56	35	26		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page